Tax Abatements for Data Centers and Government Output: Short-Run Effects on Public Sector Activity*

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State and local governments use tax abatements to attract data centers, which are capital-intensive facilities with relatively low employment. Prior research on tax incentives has documented positive and negative spillovers depending upon the duration of the firm's tenure, the size of the abatement or subsidy, and the type of firm. This paper estimates the effects of data center location on local government and sector-specific output per capita from attracting data centers through tax abatements. Using a panel dataset of 135 tax abatements granted by 44 counties in 20 states over the past 20 years, I estimate that a tax abatement for a data center led to an increase in government sector output in the year following the incentive. This increase persists into the second year, suggesting modest short-term fiscal changes to the locality. Analysis of county financial data shows no significant increase in tax collections but a marginally significant rise in debt borrowing, indicating that short-run fiscal impacts reflect accommodation costs financed through debt rather than tax base expansion.

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1. Introduction

State and local governments have increasingly turned to tax incentives as a means to spur economic development. These incentives include a combination of job creation tax credits, tax abatements, and other sorts of variation in taxes to spur job creation and private investment within their jurisdictions of governance. Among these policies, targeted tax abatements for data centers have become widespread, with 31 states and many localities enacting incentive programs.[1,2] Data centers are generally capital-intensive warehouses used for cloud storage and data processing and do not employ a substantial workforce. Unlike subsidies for manufacturing plants or headquarters relocation, the public return for granting tax incentives to data centers is less clear. This paper estimates the effects of data center location on local government and sector-specific output.

Data centers differ from the typical firms analyzed in the economic development literature due to having a relatively small workforce and requiring a large capital investment. A method of evaluating the relative size of a local economic development subsidy or tax abatement is to divide the size of the subsidy by the number of permanent jobs expected to be created by the firm, which provides an informal means of evaluating the tax incentive relative to the expected average wage created by the plant. For example, the BMW plant in Greenville-Spartanburg, SC, received an incentive of $115 million ($57,500 per expected job), and Boeing received a $50 million incentive ($100,000/job) by moving their headquarters to Chicago.[3] The rationale behind this calculation is that the average worker contributes to state and local tax revenues, so the "efficient" incentive would equal the present value of those payments minus the expected cost of public services consumed by the firm and workers. However, data centers employ few workers. As a result, the standard cost-per-job framework is inadequate for assessing the benefits of a data center's location.

This paper answers the question: what are the local sectoral effects of attracting a data center through local tax abatements? Using a panel dataset of 135 tax abatements granted by 44

[1] This is as of 2022.
[2] AP News (2022),(Mangum, 2020)
[3] (Greenstone and Moretti, 2003)

counties across 20 states over the past two decades, I estimate the effect of a data center locating on government sector output, GDP, and related industries including information technology, finance, healthcare, utilities, and construction. I find that, in the year following a data center abatement and location, government and government enterprise output increases by approximately $200 per capita. This increase persists in the second year following location. Other sectoral outcomes show no statistically significant change, suggesting that the main measurable effect occurs through changes in local public finances. These results point to a measurable fiscal increase, even in the absence of large employment gains, and suggest that data center abatements and subsequent location may contribute to local revenue indirectly through local tax base growth by increasing commercial property. However, this initial result is not conclusive; the increase could also be attributed to increased borrowing and subsequent output dedicated to public "accommodating" costs for the data center(s).

To investigate whether localities financed this increased government activity through expanded tax collections or borrowing, I collected county-level financial data from Annual Comprehensive Financial Reports (ACFRs) for 42 counties in the sample. From these reports I extracted total tax revenue, property tax collections, debt borrowing, and bonded debt for the period 2001-2020. I find no statistically significant effects on tax revenue or property tax collections in the two years following abatement, but estimate a marginally significant 5.5 percent increase in debt borrowing in the first year. These preliminary fiscal results suggest that counties may finance data center accommodation costs through borrowing rather than from an expanded tax base in the short run.

This paper is structured as follows. Section 2 provides background on data center operations, local tax bases and tax abatements, and firm location choice. Section 3 reviews the literature on economic development, and Section 4 addresses a theoretical discussion on how the abatement solves the local government's "commitment problem" to a long-term commercial property tax rate. Section 5 introduces the dataset and discusses the construction of the panel. Section 6 outlines the empirical approach, and Section 7 presents the main results. Section 8 concludes with a discussion of policy implications and directions for future research.

2. Background

This section provides institutional background on data centers and local tax abatements. I begin by outlining the general characteristics of data center facilities (Section 2.1) before examining their scale across the United States (Section 2.2). Section 2.3 explores how artificial intelligence has accelerated recent data center growth. I then analyze the factor inputs required for data center operations (Section 2.4) and discuss local employment (Section 2.5). Finally, Section 2.6 examines the structure of tax abatements and their role in firms' location decisions.

2.1 General Data Center Background

A data center is a facility that processes and stores digital information, ranging in size from small server rooms to warehouses.[4] For this paper, I focus on large warehouse-sized data centers, which often serve regional or national computing or storage needs.[5] The size of this global server market in 2022 was $83.7 billion.[6] There are three types of data centers: (1) colocation, (2) enterprise, and (3) hyperscale.[7] Colocation centers rent out server space to clients, while enterprise centers are for use by a single firm.[8] Hyperscale centers support thousands of servers and are generally for cloud computing and data storage.[9]

The capital investment reported for hyperscale data centers ranges from several hundred million to several billion dollars, but data centers employ relatively few on-site workers compared to their capital footprint.[10,11] This characteristic makes traditional metrics like cost-per-job less informative when evaluating the "cost" of tax abatements. This framing has tended to be a

[4] Mangum (2020)

[5] The Department of Energy specifies that "the larger data centers owned by major cloud providers and national supercomputer centers comprise less than 10 percent of the server market." U.S. Department of Energy (2022)

[6] Grand View Research (2022)

[7] 165 Halsey (2021)

[8] In 2021, there were 1,829 colocation data centers in the United States (Figure 1), which was the population for the earlier BEA sample. There are now over 4000 data centers in the United States.

[9] Flores (2021)

[10] This dollar amount comes from the reported capital investment numbers by Good Jobs First in the sample discussed in the data section. I present the reason why these numbers are potentially inaccurate.

[11] The median number of workers per data center in the sample in Section 5 is 50.

useful metric for evaluating the costs and benefits of attracting a more labor-intensive firm. Part of the rationale behind attracting data centers might be the possible spillover effect increases in each location, such as increased employment in financial services, information technology, and healthcare services.[12] This prompts the motivation to examine if attracting a data center by a locality actually has a positive impact on those industries.

2.2 Scale of Data Centers in the United States

The majority of hyperscale data centers are owned by a few firms, such as Amazon, Google, Meta, and Microsoft, and take about two years to build.[13] While colocation data centers are more evenly distributed nationwide, hyperscale and enterprise data centers reflect large centralized investments. The trend towards hyperscale data centers started around the time Amazon began the development of cloud-based web services in 2002.[14] Given the recent origin and the known numbers of colocation and hyperscale data centers being around 2,000 in 2020, on average, about 100 data centers were built annually between 2000-2020. However, this trend rapidly accelerated, and there are now around 4,000 data centers in the US.[15]

Since 2021, demand for data centers has increased exponentially both domestically and globally. JLL's Global Data Center Outlook found that the global market size for colocation data centers is forecasted to grow 11.3 percent from 2021-2026. The fastest-growing data center types are expected to be hyperscale or edge (smaller facilities that process data near users to minimize delays) facilities. Hyperscale data centers in particular are expected to grow by 20 percent between 2021 and 2026. Overall, the demand for data centers in the U.S. is stronger compared to other regions and has accounted for 52 percent of all data center transactions between 2018 and 2022. As of 2023, the United States controls 53 percent of global hyperscale data center capacity.[16]

[12] Expert Market Research (2022).
[13] Data Center Knowledge (2019).
[14] SiliconANGLE (2014).
[15] Data Center Map (2024)
[16] Beyler, Ryan, 2023. "Hyperscale and Edge Drive Data Center Demand as the Rise of AI Takes Center Stage." JLL, Chicago, IL, https://www.jll.com/en-us/newsroom/data-centers-2023-global-outlook

2.3 AI's Impact on Data Center Growth

The 2020s AI boom that accelerated with generative tools like OpenAI's ChatGPT, Anthropic's Claude AI, Google's Gemini, and X's Grok has spurred substantial data center growth. As more advanced large language models are continually in development, they require bigger and more powerful data centers built for AI capabilities. Research conducted by Goldman Sachs has found that as of 2025, over 50 million square feet of data center construction for AI-specific activities is in development (Min, 2025). The investment bank reports AI server racks alone "consume 10x more power than Cloud equivalents" and its greater impact will reach beyond just the market for datacenters and into a variety of sectors, especially energy and financials (Min, 2025). In conjunction with these developments, it is projected half of all cloud data centers will use AI by 2025.[17] As AI continues to create and transform industries, data centers remain the capital that is conducive to its success.

2.4 Factor Inputs for Data Center Operations

As demand for data centers has increased, so too has demand for data center factor inputs, especially electrical power and water. A December 2024 report by the U.S. Department of Energy found that data center load growth (the growth in electrical power consumed by such facilities) is expected to double or triple by 2028. Data centers consumed about 4.4 percent of total US electricity in 2023 and are projected to account for between 6.7 and 12 percent of total domestic energy consumption by 2028. Because of this phenomenon, the Department of Energy expects electricity demand in the U.S. to increase, with data center expansion and the rise of AI applications largely causing that increase. The report also found that total data center electricity usage climbed from 58 terawatt-hours (TWh) in 2014 to 176 TWh in 2023 and estimates that by 2028 the level of usage will increase to between 325 and 580 TWh by 2028.[18] Data center operations also require

[17] Beyler, Ryan, 2023. "Hyperscale and Edge Drive Data Center Demand as the Rise of AI Takes Center Stage." JLL, Chicago, IL, https://www.jll.com/en-us/newsroom/data-centers-2023-global-outlook

[18] U.S. Department of Energy, 2024. " DOE Releases New Report Evaluating Increase in Electricity Demand from Data Centers." U.S. Department of Energy, Washington, DC, https://www.energy.gov/articles/doe-releases-new-

vast amounts of water consumption, which is used for electrical equipment air coolant. Although the true level of water consumed by data centers is unknown because most of them consume from municipal utilities, the cooling process is believed to use millions of gallons of water annually.[19]

2.5 Labor Implications of Data Centers

Although data center construction has increased 69 percent year over year from 2023 to 2024 and is currently at an all-time high, growth in data center employment has been less dramatic. The U.S. Bureau of Labor Statistics found that although the number of people working in data centers grew from 306,000 in 2016 to 501,000 in 2023, the Quarterly Workforce Indicators (QWI) report conducted by the U.S. Census Bureau shows that employment growth is concentrated.[20] Five states account for over 40 percent of U.S. data center employment: California, Texas, Florida, New York, and Georgia, with California and Texas jointly accounting for 27 percent. Nonetheless, despite the presently strong data center demand, there remains a skill shortage of high-tech professionals that may test the data center market's recent growth. As of 2023, younger workers lacking the required technical skills have presented a challenge for data center growth to maintain its pace.[21] In response, both the private and government sectors have taken the initiative to promote a robust data center employment pipeline. Most notably, the CHIPS Act of 2022 offered grants and loans with the goals of strengthening technology education, including in the realm of data centers.

2.6 Local Taxes Abated and Location Decision by the Firm

The most common rebates come from reductions in (1) State and Local Sales and Use Taxes, (2) Real (unmovable) Property Taxes, and (3) Personal (movable) Property Taxes.[22] Among

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[19]Ta, Linh, and Alissa Widman Neese, 2025. "The Midwest's Data Center Boom is Heating Up." Axios, Arlington, VA, https://www.axios.com/2025/04/16/midwest-data-center-growth-energy-usage

[20]Foote, Andrew, and Caelan Wilkie-Rogers, "Employment in Data Centers Increased by More Than 60 percent From 2016 to 2023 But Growth Was Uneven Across the United States" U.S. Census Bureau, Suitland, MD, https://www.census.gov/library/stories/2025/01/data-centers.html

[21]Beyler, Ryan, 2023. "Hyperscale and Edge Drive Data Center Demand as the Rise of AI Takes Center Stage." JLL, Chicago, IL, https://www.jll.com/en-us/newsroom/data-centers-2023-global-outlook

[22]Tarczynska (2016)

these, the most relevant for these abatements are commercial property taxes, particularly in rural areas where a single data center could substantially increase the property tax base if it consists of improvements to the site value of the land. That increase in the commercial property tax base could justify the abatement in net present value, especially if it contributes to broader local economic activity, including increases in government sector output. If the abatement is just functioning as a means of getting the locality to commit to a valuation method and rate for a fixed period of time, it would substantially reframe the analysis of these as subsidies. This context also underpins the commitment problem: if firms anticipate future tax hikes without an abatement agreement in place, they may opt to locate in jurisdictions that commit to a stable tax rate and valuation method.

Firms likely select counties based in part on their commitment to a stable tax regime, alongside other characteristics like energy costs and climate risk. The counties are not selected at random from all counties in the United States. Rather, companies look for the following characteristics in a location: relatively low energy costs for running the equipment and keeping it from overheating, low risk of significant natural disasters that could put the center off-line, and distance from other data centers to spread the risk of a disaster taking multiple facilities off-line. Further, they may be influenced by whether the state has a tax abatement or local economic development program specifically for data centers. For the purpose of this paper, I will focus on the presence of incentive programs and commitment to a stable tax regime.

3. Literature Review

This paper contributes to the tax incentives and economic development literature. Tiebout's (1956) framing of local governments as competing providers of public goods, with individuals sorting across jurisdictions, according to their preferences, lays the foundation for understanding how localities may compete to attract firms. Willis (1985) finds that local investment subsidies may be justified on a cost-benefit basis depending on the length of time a firm is present, which is a key determinant of whether the net present value of the subsidy for that development is positive.

In one of the earliest formal treatments, *Bidding for Firms*, Black and Hoyt (1989) show how tax incentives emerge in equilibrium as jurisdictions compete for mobile firms. Glaeser (2001) acknowledges that the development incentives seem to be a permanent part of the urban economic landscape. Greenstone and Moretti (2003) use matched pairs of "winning" and "losing" counties from corporate location decisions to estimate the effects of tax subsidies on local welfare. They find that winning counties experience a 1.5 percent annual increase in wage bills and a 1.1 percent annual increase in property values, with no evidence of reduced local government funding.

Agrawal, Hoyt, and Wilson (2022) provide a survey of the literature on local tax competition and firm bidding. Slattery and Zidar (2020) find that state and local business tax incentives may lead to direct employment gains within targeted industries, but do not find strong evidence of broader state or local economic growth. Bidding for firms has generated mixed findings in the literature, with some estimating positive effects on local outcomes and others finding a race-to-the-bottom: (Doyle and van Wijnbergen, 1994), (Mast, 2020), (Ossa, 2015), (Wassmer and Anderson, 2001), (Kang et al., 2016), (Wilson and Wildasin, 2004). In the most recent contribution to this literature, Slattery (2025) finds that most of the surplus is captured by firms and casts doubt on the effectiveness of subsidies as place-based development policy.

In regards to studies on the effect of abatements specifically, in 1974, the state of Michigan adopted the "Industrial Facilities Tax" abatement program to spur industrial development by allowing localities to abate taxes. Using data from that program, Anderson and Wassmer (2001) find that specific forms of local incentives exert a positive influence on local and commercial property value. Skidmore, Kang, and Reese (2016) find positive spillovers on residential and commercial property growth, but these spillovers are smaller than the "foregone" tax revenue. There has also been related research on the public subsidization of stadiums and the effect of a sports franchise's location on earnings and employment (Coates, 2003) and on employment and wages (Jasina and Rotthoff, 2008).

4. Theory

A plausible motivation behind offering a tax abatement to data centers is that it solves a commitment problem on the part of the local government. A pre-existing tax agreement, by definition, includes a term length for an agreed-upon rate and valuation method for that commercial property. In a two-period setting, if a firm were to invest in building a data center without an agreement, the local government could *renege* on expected future tax liability forecast by the firm by raising commercial property tax rates and by increasing the *de minimis* exemption for residential property. Even limited knowledge of local elections could create conceivable circumstances in which a firm would not want to expend capital without this agreement. Since the typical data center investment in this sample ranges from several hundred million to several billion dollars in fixed capital, the potential commercial tax liability, without an agreed-upon rate and valuation method, is substantial. Without a binding commitment, firms anticipate the possibility of future rate hikes once immobile capital is in place and, therefore, may choose to locate in jurisdictions that commit to a stable tax regime via tax abatement.

A negotiated tax abatement offers commitment from the locality *ex-ante* and, therefore, is likely necessary for attracting capital investment. The empirical results in section 7 suggest that this commitment is not a "zero-sum" game on the part of the locality. They stand to gain potential government expenditure via an increased commercial property tax base, even if it is at an agreed-upon lower rate. In this light, the abatement, which the data compiler treats as equivalent to a "subsidy", is not a subsidy in the technical use or meaning of the word. The abatements act as credible commitment mechanisms that have the potential to expand the tax base of local governments.

4.1 Model Setup

The purpose of this model is to formalize the commitment problem facing local governments and firms in the data center investment decisions. I demonstrate that tax abatements serve as

credible commitment devices solving time-inconsistency problems, leading to Pareto-improving outcomes even under complete information. This two-stage framework follows the constitutional economics approach of Buchanan and Tullock (1962), where institutional rules are established *ex-ante* under uncertainty to govern subsequent political decisions.

4.2 Players and Setting

I consider a strategic interaction between two players: a local government, denoted by G, and a firm, denoted by F. The local government controls commercial property tax policy, while the firm makes an irreversible capital investment decision regarding data center location. The firm considers whether to invest in building a data center, which requires substantial capital investment. The government seeks to maximize tax revenue net of the public infrastructure costs necessary to accommodate the facility.

4.3 Government Objective and Revenue Constraint

The government faces a revenue requirement $\bar{R} > 0$ to finance public goods and services, and must pay accommodation costs $K > 0$ if the data center locates, including roads, utilities, and other public infrastructure. The government can tax two bases: the existing residential property base with assessed value $V_R > 0$ at rate τ_R, and commercial property (the data center) with assessed value $V > 0$ at rate τ_C.

The government's budget constraint requires that total tax revenue from both bases, net of accommodation costs, equals the revenue target:

$$\tau_R \cdot V_R + \tau_C \cdot V - K = \bar{R}. \tag{1}$$

The government's objective is to minimize the tax burden on residential property owners.[23] Using the budget constraint to solve for the residential tax rate, we obtain $\tau_R = (\bar{R} - \tau_C \cdot V + K)/V_R$.

[23] This is a rationale assumption if local politicians are to get elected. Residential taxpayers in nearly all circumstances will provide more voters than commercial taxpayers.

Therefore, minimizing τ_R is equivalent to maximizing the commercial tax base contribution $\tau_C \cdot V - K$. This formulation captures a key political economy insight: local governments seek to protect residential taxpayers by shifting the tax burden onto commercial property when possible, expecially when that commercial property does not employ that many residential taxpayers. This objective creates a strong *ex-post* incentive to extract maximum revenue from immobile commercial capital once it is in place. Additionally, this formulation also incorporates the reality that data centers will not have a lot of residential property taxpayers, so there are few residential taxpayers the government needs to satisfy.

4.4 Timing and Institutional Regimes

The game proceeds in two periods, preceded by an *ex-ante* institutional stage that determines the government's commitment capacity. Following Buchanan and Tullock (1962), this *ex-ante* stage represents a constitutional choice made before specific investment opportunities arise, analogous to their distinction between constitutional and post-constitutional decision-making. At this *ex-ante* stage, the institutional regime is established. The government operates under two possible regimes: Regime A, in which a tax abatement institution exists and the government can credibly commit to future tax rates, or Regime N, in which no such commitment technology is available and the government retains full control over future tax policy.

In Period 1, the firm observes the institutional regime and decides whether to invest. The firm's choice set is denoted $a_F \in \{I, NI\}$, where I represents the decision to invest in building the data center and NI represents the decision not to invest. This investment decision is irreversible; once capital is sunk into the facility, it cannot be redeployed.

In Period 2, after observing the firm's investment decision, the government sets the commercial property tax rate τ_C. The government's choice set depends on the institutional regime established *ex-ante*. Under Regime N, the government faces no constraints and may set any tax rate within the feasible range $\tau_C \in [\tau_{\min}, \tau_{\max}]$, where $\tau_{\min}$ represents the minimum politically or legally feasible rate and $\tau_{\max}$ represents the maximum rate. Under Regime A, the government is bound by its

ex-ante commitment and must set $\tau_C = \bar{\tau}$, where $\bar{\tau} < \tau_{\max}$ is the rate specified in the binding tax abatement agreement.

4.5 Payoff Structure

The firm's two-period payoff depends on its investment decision and the commercial tax rate. If the firm invests, it receives gross revenue $R > 0$ from data center operations in Period 2, pays property taxes based on the assessed value $V > 0$ at rate τ_C, and incurs fixed capital investment costs $F > 0$ in Period 1 and operating costs $C > 0$ in Period 2. The firm's payoff function is given by:

$$U_F = \begin{cases} R - \tau_C \cdot V - F - C & \text{if } a_F = I \\ 0 & \text{if } a_F = NI. \end{cases} \tag{2}$$

The government's payoff reflects its objective to minimize the residential tax burden. If the firm does not invest, the government must raise the entire revenue requirement $\bar{R}$ from residential taxpayers, resulting in $\tau_R = \bar{R}/V_R$. If the firm invests, the commercial tax base contribution allows the government to reduce the residential burden to $\tau_R = (\bar{R} - \tau_C \cdot V + K)/V_R$. The government's payoff function can be expressed as the negative of the residential tax rate:

$$U_G = \begin{cases} -\tau_R = -\frac{\bar{R} - \tau_C \cdot V + K}{V_R} & \text{if } a_F = I \\ -\frac{\bar{R}}{V_R} & \text{if } a_F = NI \end{cases} \tag{3}$$

where $\bar{R} > 0$ is the government's revenue requirement, $K > 0$ represents accommodation costs for the data center, and $V_R > 0$ is the assessed value of the residential property tax base. This payoff structure shows why a government prefers higher commercial tax rates when the firm has invested, as each additional dollar of commercial tax revenue reduces the residential burden by $1/V_R$ dollars per capita.

4.6 Assumptions

The model relies on four key assumptions that capture the hold-up problem and ensure the existence of mutually beneficial investment opportunities. First, I assume that investment is viable at the minimum possible commercial tax rate, such that:

$$R - \tau_{\min} \cdot V - F - C > 0. \tag{4}$$

This ensures that under favorable tax conditions, the investment generates positive returns for the firm. The gross revenue of the firm less the commercial tax liability, fixed investment costs, and operating costs is greater than zero.

Second, I assume the hold-up problem is severe: investment is unprofitable at the maximum commercial tax rate, meaning:

$$R - \tau_{\max} \cdot V - F - C < 0. \tag{5}$$

This assumption captures the time-inconsistency problem central to the theory. The firm cannot profitably invest if it anticipates facing the maximum tax rate *ex-post*.

Third, the government's *ex-post* incentive to shift the tax burden is substantial. Once the data center is in place with immobile capital, setting $\tau_C = \tau_{\max}$ allows the government to reduce the residential tax burden compared to the no-investment scenario, reflected in the condition:

$$\tau_{\max} \cdot V - K > 0. \tag{6}$$

This inequality ensures that the commercial tax base contribution exceeds accommodation costs, making the government prefer to extract maximum revenue from the data center to protect residential taxpayers.

Fourth, a Pareto-improving committed rate exists: there exists some $\bar{\tau} \in (\tau_{\min}, \tau_{\max})$ satisfy-

ing both:

$$R - \bar{\tau} \cdot V - F - C > 0 \quad \text{and} \quad \bar{\tau} \cdot V - K > 0. \tag{7}$$

This ensures that a binding commitment can make both the firm and residential taxpayers strictly better off than the no-investment outcome.

Lastly, I assume complete information: all payoff parameters $(R, V, F, C, K, V_R, \bar{R}, \tau_{\min}, \tau_{\max}, \bar{\tau})$, the timing structure, and each player's rationality are common knowledge. This assumption ensures that the commitment problem arises purely from time inconsistency rather than information asymmetry. While complete information is admittedly a simplification, it serves as a natural baseline for isolating the hold-up problem and provides a benchmark against which to evaluate the welfare gains from commitment mechanisms. Moreover, for large-scale data center investments involving extensive negotiations and due diligence, the assumption is more plausible than for smaller-scale projects: key parameters such as investment costs, property valuations, and infrastructure requirements are typically disclosed or can be credibly estimated by both parties during abatement negotiations.

4.7 Equilibrium Analysis

Backward induction is used to solve for the Subgame Perfect Nash Equilibrium (SPNE) under both institutional regimes.

4.7.1 Regime N: No Abatement Institution

In Period 2, if the firm has invested, the government chooses the commercial tax rate τ_C to minimize the residential tax burden τ_R. Using the budget constraint $\tau_R = (\bar{R} - \tau_C \cdot V + K)/V_R$, minimizing τ_R is equivalent to maximizing $\tau_C \cdot V - K$ over $\tau_C \in [\tau_{\min}, \tau_{\max}]$. Since this objective is linear and increasing in τ_C, the government sets $\tau_C^N(I) = \tau_{\max}$. This maximizes the commercial base's contribution to the revenue target, thereby minimizing the burden on residential taxpayers.

In Period 1, anticipating this behavior, the firm compares the payoff from investing, $U_F(I) = R - \tau_{\max} \cdot V - F - C < 0$ (by Assumption 2 in equation 5), with the payoff from not investing,

$U_F(NI) = 0$. Because investment yields negative returns when facing the maximum tax rate, the firm rationally chooses not to invest. The unique SPNE, or No Investment Equilibrium, under Regime N is $(s_F^N, s_C^N) = (NI, \tau_{\max})$ with payoffs $(U_F^N, U_G^N) = (0, -\bar{R}/V_R)$. In plain terms, the firm chooses not to invest (NI) because it anticipates the government will set the commercial tax rate at the maximum ($\tau_{\max}$), leaving the firm with a payoff of 0 and forcing the government to collect all revenue from residential taxpayers (a burden of $\bar{R}/V_R$).

4.7.2 Regime A: Tax Abatement Institution

Under Regime A, the government is constitutionally bound to the committed rate $\tau_C^A = \bar{\tau}$. The government cannot deviate from this rate in Period 2, regardless of the firm's investment decision.

In Period 1, knowing with certainty that the rate will be $\bar{\tau}$, the firm compares $U_F(I) = R - \bar{\tau} \cdot V - F - C > 0$ (by Assumption 4 in equation 7) with $U_F(NI) = 0$. Since the committed rate makes investment profitable, the firm invests. The unique SPNE, or Investment Equilibrium with an Abatement, under Regime A is $(s_F^A, s_C^A) = (I, \bar{\tau})$ with payoffs:

$$(U_F^A, U_G^A) = \left(R - \bar{\tau} \cdot V - F - C, -\frac{\bar{R} - \bar{\tau} \cdot V + K}{V_R} \right) \tag{8}$$

where the firm earns positive profit and the residential tax burden is strictly lower than under Regime N.

If $\bar{\tau} \in (\tau_{\min}, \tau_{\max})$ satisfies both conditions in equation (7), then Regime A strictly Pareto dominates Regime N: $U_F^A > U_F^N = 0$ and $U_G^A > U_G^N$, as the residential tax burden falls from $\bar{R}/V_R$ to $(\bar{R} - \bar{\tau} \cdot V + K)/V_R < \bar{R}/V_R$. In other words, both parties are strictly better off with the abatement: the firm earns positive profit ($U_F^A > 0$) instead of staying out, and residential taxpayers enjoy a lower tax burden ($U_G^A > U_G^N$) because the data center's tax payments partially offset the accommodation costs.

4.7.3 Interpretation

The model formalizes the commitment problem that motivates tax abatement agreements for data center investments. Without a binding agreement, the local government would set $\tau_C = \tau_{\max}$ in Period 2 to minimize the residential tax burden, not out of opportunism but to protect residential taxpayers from bearing the full revenue requirement. This creates a time-inconsistency problem: firms anticipate facing the maximum rate once immobile capital is in place and, recognizing that $R - \tau_{\max} \cdot V - F - C < 0$, rationally choose not to invest. A negotiated tax abatement offers credible commitment *ex-ante*, enabling the firm to invest profitably at the committed rate $\bar{\tau}$. The equilibrium demonstrates this is not a "zero-sum" game—the locality gains by expanding the commercial tax base, which reduces the residential burden from $\bar{R}/V_R$ to $(\bar{R} - \bar{\tau} \cdot V + K)/V_R$ even at the lower committed rate. The abatement is not a subsidy but rather a constitutional utilitarian contract in the sense of Buchanan and Tullock (1962), establishing binding rules *ex-ante* that prevent *ex-post* burden-shifting and enable Pareto-improving investment.

4.8 Comparative Statics

As fixed investment F increases, the derivative of firm utility with respect to fixed investment, $\partial U_F^A / \partial F = -1 < 0$, indicates the firm becomes more vulnerable to the hold-up problem, making commitment more valuable. This explains why data centers, with substantial capital investments and limited labor to lobby on their behalf to keep rates low, benefits more from abatements compared to smaller commercial investments. As assessed value V increases, the government's incentive to shift the tax burden onto the commercial base increases, since higher property values represent a larger potential contribution to meeting the revenue target $\bar{R}$. However, under commitment, the derivative of government utility with respect to assessed value, $\partial U_G^A / \partial V = \bar{\tau}/V_R > 0$, shows that higher property values increase the commercial contribution to the revenue target, as each additional dollar of commercial value assessed at rate $\bar{\tau}$ reduces the residential rate by $\bar{\tau}/V_R$ dollars per unit of residential property value. The optimal abatement rate from the government's perspective, subject

to ensuring firm participation, $\bar{\tau}^* = (R - F - C)/V$, which maximizes the commercial contribution while making the firm indifferent between investing and not investing. This rate extracts the feasible tax revenue from the commercial base while still inducing investment.

4.9 Testable Implications

The model generates empirically testable predictions consistent with observed data center location and fiscal outcomes. First, data centers should locate in jurisdictions with abatement institutions rather than those without commitment mechanisms:

$$\Pr(\text{Invest} \mid \text{Regime A}) > \Pr(\text{Invest} \mid \text{Regime N}) \tag{9}$$

Second, commercial property tax revenue should increase post-abatement: $\Delta T^{A}_{\text{commercial}} = \bar{\tau} \cdot V > 0$. The commercial base expands from zero to V, allowing the locality to meet its revenue target $\bar{R}$ while reducing the residential tax burden. Third, total government revenue should increase as the commercial base expands, enabling lower residential tax rates. The net fiscal benefit, $\bar{\tau} \cdot V - K > 0$, represents the commercial contribution beyond accommodation costs, as the expanded tax base allows the locality to meet revenue obligations while reducing the per-capita residential burden from $\bar{R}/V_R$ to $(\bar{R} - \bar{\tau} \cdot V + K)/V_R$. In the empirical analysis that follows, I focus on testing the second and third predictions using county-level fiscal data, deferring a test of the first prediction to a future draft.

The empirical results in Section 7 provide a preliminary test of these fiscal predictions. The BEA results show approximately $200 per capita increases in government sector output following abatement, but county-level financial data reveal no significant increases in tax revenue or property tax collections, and instead a marginally significant 5.5 percent increase in debt borrowing in the first year. These patterns suggest that accommodation costs K may exceed the immediate commercial tax contribution $\bar{\tau} \cdot V$ in the short run, with the model's predicted fiscal benefits potentially occurring over longer time horizons as data centers become fully operational and assessed.

5. Data

In Subsection 5.1, I present the creation of the tax abatement sample, and in Subsection 5.2, I discuss economic outcomes of interest for the analysis.

5.1 Tax Abatement Sample

To construct the sample of data centers that received a tax abatement, I started from a dataset of tax abatements and local development subsidies compiled by the organization *Good Jobs First.* The organization created a "subsidy-tracker," which is a data scraper that web-scrapes all publicly available local, state, and federal tax incentives. Those "events" are then categorized by keywords and industries. While numeric values reported in the subsidy tracker should be interpreted with caution due to the automated nature of data collection, the events themselves are generally reliable and well-documented.[24]

Using this dataset, in September 2021, I searched for all events related to "data centers" and found 290 results.[25] I eliminated all events that didn't have state, county, or city, and year information, because I would not be able to triangulate the county location without that information. For eight companies, localities granted multiple tax abatements in the same year. They were reported separately either because the abatement consisted of a different type of tax or the company received a "follow-up" abatement within the same year. Due to these follow-ups, especially for the abatements in Washington and New York, a potential concern is that the abatements may be for the same data center or different data center projects for the same company. I consolidated each of

[24]The tracker uses automated web scraping to extract data from publicly available sources. In a random audit of approximately twenty events, it appeared that numeric values may sometimes be associated with nearby keywords rather than the intended variable (e.g., a $20 million figure near the word "capital" seems to be coded as capital investment regardless of actual context in some cases). However, sources for each entry are posted and typically traceable with time investment. Despite these apparent limitations with numeric precision, the tracker provides a valuable standardized foundation for identifying data center-related abatements nationwide. The sample includes colocation, enterprise, and hyperscale data centers.

[25]If the tracker is accurate, this total of 290 out of an estimated 2,000 colocation and hyperscale centers nationwide. Assuming around 100 are built each year since 2000, that means about 29 percent of new data centers get some kind of tax incentive.

those abatements into one event, and the remaining sample had 135 abatement events.

Table 1: Summary of Abatements in the Sample

Category	**Count**
Overall Totals	
Total States with Abatements	20
Total Counties Treated	44
Total Abatements	135
Distribution of Abatements Across Counties	
Counties with 1 Abatement	28
Counties with 2 Abatements	5
Counties with 3 Abatements	3
Counties with 4–5 Abatements	3
Counties with 9–18 Abatements	5
Abatements in Top 4 States	
Total-Abatements in Top 4 States	100
– New York (Erie 16, Rockland 14)	35
– Oregon (Washington 4)	9
– Virginia (Loudoun 18, Henrico 3)	28
– Washington (Grant 14, Douglas 9, King 5)	28

Note: This table summarizes the number and distribution of tax abatements across states and counties in the sample.

Table 1 summarizes the distribution of data centers in the sample by county and highlights concentration in four states.[26] Figures 1 and 2 provide context on how representative the sample is relative to the national distribution of data centers. Figure 1 shows the location of 1,829 data centers across the United States, while Figure 2 shows the subset included in the sample.[27] Even though these are comparing types of data centers, the comparison suggests that California and Texas are underrepresented in the sample, while New York and Virginia are overrepresented. Though, the Erie county data centers likely located there for relatively low cost electricity and the data centers in Loudon county are there to support the Washington D.C. security apparatus. It is not clear whether this distribution is driven by the incentives for data centers, the type of data center, the quality of

[26]Table 8 in the back gives the full state and county distribution for all abatements in the sample.

[27]The temporal distribution of abatements is shown in Table 10, with notable clustering between 2011-2014.

local reporting to be scraped on data center incentives, or the bias within the subsidy tracker.[28]

5.2 Bureau of Economic Analysis and Census Data

In order to construct the outcome variables, I match the tax abatement event data to annual county-level economic data from the Bureau of Economic Analysis (BEA) and county population data from the Census Bureau. For the years 2001-2020, I collect GDP data for seven sectors: (1) total GDP, (2) financial services, (3) information technology, (4) government and government enterprises, (5) healthcare and social assistance, (6) utilities, and (7) construction. I chose these sectors based on different outcomes it seemed a data center would impact the most substantially. Financial services and information technology are industries that seem to have the most obvious benefit from access to improved digital infrastructure. I include healthcare services because some of the given industry codes in the sample were for healthcare companies. I include utilities because of the high energy consumption required to operate data centers. Construction is included to capture the short-run effects of building the facility, which could be significant in rural counties.

The government and government enterprises sector computed by the BEA is a measure of government production. This output is computed as the sum of employee compensation, consumption of fixed capital and intermediate goods and services purchased.[29] It is composed of activity by federal, state, and local governments combined. While it could be inferred from the county level and timing of the events that this metric is local public expenditures, it is most accurate to interpret this measure as a change in government output, not a direct measure of local public expenditures.[30]

In order to account for differences in county size, I create a per capita measure of GDP using population information from the Census Bureau. Using this per-capita sector-specific unit allows for the estimation of the changes within each county over time. These sector-specific GDP per capita values were matched to the abatements by county, state, and year to form the panel used for

[28]Good Jobs First "is a national policy resource center for grassroots groups and public officials, promoting corporate and government accountability in economic development and smart growth for working families... The Project is designed to be a resource to aid activism." Quote from the organization's website. Good Jobs First (2022)

[29]Bureau of Economic Analysis (2023), *NIPA Handbook, Chapter 9*, p. 9-2.

[30]Summary statistics for the BEA regression variables are presented in Table 11.

analysis. This panel forms the basis for all regression outcomes presented in Section 7.

5.3 County-Level Financial Data

To construct county-level fiscal outcome variables, I collect financial data from Annual Comprehensive Financial Reports (ACFRs). ACFRs are standardized financial statements that local governments produce annually to report their financial position, results of operations, and cash flows in accordance with Generally Accepted Accounting Principles (GAAP). These reports provide detailed information on revenues, expenditures, assets, liabilities, and debt obligations at the county level. For counties in the tax abatement sample, I obtained ACFRs by downloading publicly available reports from county government websites or, when reports were not readily accessible online, by submitting Freedom of Information Act (FOIA) requests directly to county finance departments.

From these ACFRs, I extract four fiscal outcome variables for the years 2001-2020: total tax revenue, property tax collections, debt borrowing, and bonded debt.[31] Total tax revenue represents the sum of all tax collections by the county government, including property taxes, sales taxes, and other local taxes. Property tax collections captures revenues from *ad valorem* taxes on real and personal property, which is the primary revenue source directly affected by data center tax abatements. Debt borrowing is a broad measure that includes all forms of county borrowing, encompassing municipal bonds, short-term notes, bank loans, lines of credit, and other debt instruments. Bonded debt is a narrower measure that captures only formal, long-term municipal bonds issued in public bond markets, representing a subset of total debt borrowing. The distinction between these two debt measures reflects differences in borrowing mechanisms—bonded debt requires formal bond issuance procedures and is typically used by larger counties, while debt borrowing includes more flexible and informal borrowing arrangements accessible to smaller jurisdictions.

The availability of these fiscal variables varies substantially across counties in the sample.[32] For

[31] Descriptive statistics for the ACFR fiscal outcome variables are shown in Table 13.

[32] Table 12 shows the variation in data coverage across outcome variables, ranging from 87 percent for total tax

total tax revenue and property tax collections, data are available for 42 and 41 counties respectively, representing approximately 85-87 percent of the 44 counties with data center abatements in the BEA analysis sample. However, debt borrowing data is available for only 23 counties (53 percent), and bonded debt data is available for only 16 counties (42 percent). This attrition in the debt variables occurs because smaller counties often do not report detailed debt information in their ACFRs, either because they do not issue formal bonds or because their financial reporting is less comprehensive. Counties that do report detailed debt information tend to be larger jurisdictions with more sophisticated financial management systems and greater borrowing activity. This sample selection may introduce bias if counties that report debt data differ systematically from those that do not in ways that affect their response to data center abatements.

Following the same approach used for the BEA analysis, I structure the data at the company-event level rather than the county level. Because multiple companies can receive tax abatements in the same county across different years, each company's abatement event is treated as a separate observation with a unique identifier (CompanyCountyEvent_ID). All companies receiving abatements in the same county share the same county-level financial outcomes, as these variables reflect aggregate county finances rather than company-specific effects. This structure allows for separate treatment effects to be estimated for each abatement event while accounting for the fact that multiple events in the same county are not independent observations. The fiscal data are matched to abatement events by county, state, and year to form a balanced panel for the 2001-2020 period. This panel forms the basis for the regression analysis of fiscal outcomes presented in Section 7.2.

The collection of ACFR data is ongoing, and the sample will expand as additional counties provide financial reports or as FOIA requests are fulfilled. As more comprehensive county-level financial data become available, the results presented in Section 7.2 should be interpreted as preliminary estimates that will be refined with expanded data coverage.

revenue to 42 percent for bonded debt.

6. Empirical Approach

I use a two-way fixed effects (TWFE) regression to estimate the short-run effects of data center location on sectoral outcomes. Each county in the sample receives at least one data center tax abatement, with some receiving many abatements over time. In two separate regressions for the same outcome of interest, I estimate the effect of a data center abatement in the first and second year following implementation. For each specification, I create a treatment variable that equals 1 only in the year of interest (either the first or second year post-abatement), and 0 in all other years. This allows me to estimate short-run effects without imposing assumptions about trends, duration, or persistence beyond this immediate period. I estimate the following regression:

$$y_{it} = \beta \cdot \text{Post}_{it} + \alpha_i + \delta_t + \varepsilon_{it}. \tag{1}$$

The dependent variable, y_{it}, is the specific sectoral outcome (GDP, government, finance, information technology, construction, healthcare, utilities, total tax collections, property tax collections, debt borrowing, or debt bonded). The Post_{it} is the indicator that equals 1 only in year post-abatement or two years post-abatement in a separate regression. The county and time-fixed effects are captured by α_i and δ_t, respectively. The coefficient of interest, β, captures the average variation in the dependent variable with respect to receiving the abatement in the specified year.[33] For the BEA sectoral outcomes (GDP, government, finance, information technology, construction, healthcare, and utilities), α_i represents county fixed effects. For the county-level fiscal outcomes (total tax collections, property tax collections, debt borrowing, and bonded debt), α_i represents Company-CountyEvent_ID fixed effects, which assign a unique identifier to each company's abatement event, allowing multiple events within the same county to be treated separately while sharing the same county-level fiscal outcomes.

[33] In future work, I plan to explore empirical approaches that account for selection into the treatment, similar to the winners and losers control groups used by Greenstone and Moretti (2003). Alternatively, instruments based on local characteristics like weather or energy costs may isolate the variation from the data center location. Lastly, I would add controls for state-level tax incentives.

7. Results

In section 7.1, I present the Bureau of Economic Analysis results, and in section 7.2, I present the Annual Comprehensive Financial Reports results.

7.1 Bureau of Economic Analysis GDP Results

Using the TWFE regression from equation (1), I estimate the effects of data center tax abatements on per capita GDP and sector-specific economic activity. Across the finance, information technology, healthcare, utilities, and construction sectors and overall GDP, I find no statistically significant effects in the year after abatement or two years after. I also estimate a separate regression for construction activity in the year of the abatement, in case projects were "shovel-ready" and the granted abatement could explain some immediate variance within construction. This estimate is also insignificant. These insignificant results suggest that data center location post-abatement does not generate detectable short-run changes in economic activity across these sectors.

The two significant results are in the government and government enterprises sector. I estimate that per capita government sector output increased by $198.56 per person. This result is marginally significant, assuming homoskedastic standard errors. The estimate remains the same after county-clustered standard errors, which control for autocorrelation in a county over time and differences within error size across counties (heteroskedasticity) (Table 2), but this result does not remain significant (Table 3).[34] I estimate that per capita government sector output remains positive in the two years after the abatement with an estimated increase of $173.29 per person.[35]

[34]The difference between homoskedastic and clustered standard errors is particularly important here because the sample comprises 44 distinct county-level treatment events rather than one large pooled sample. This limited cross-sectional variation means that unexplained heterogeneity across counties affects standard errors when properly accounting for within-county correlation.

[35]Summary statistics in Table 3.

Table 2: Comparison of First-Year Government/Capita Estimates

Model	**Coefficient**	**Std. Error**	**t-Statistic**	**Adj.** R^2
TWFE (Clustered SEs)	0.19856	0.13294	1.49364	0.821
TWFE (Homoskedastic SEs)	0.19856**	0.09350	2.12400	0.821

Note: The dependent variable is GDP for government sector output per capita in thousands of dollars. Both models include county and year-fixed effects. The first row uses standard errors clustered at the county level, while the second assumes homoskedasticity. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 3: Comparison of Second-Year Government/Capita Effect Estimates

Model	**Coefficient**	**Std. Error**	**t-Statistic**	**Adj.** R^2
TWFE (Clustered SEs)	0.17329	0.14045	1.23384	0.820
TWFE (Homoskedastic SEs)	0.17329*	0.09469	1.83000	0.820

Note: The dependent variable is GDP for government sector output per capita in thousands of dollars. Both models include county and year-fixed effects. The first row uses standard errors clustered at the county level, while the second assumes homoskedasticity. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The clustered standard errors provide a more conservative assessment of the effects of the policy, but the point estimates provide suggestive evidence that the government sector output is not decreasing after attracting a data center. The government GDP per capita is plotted over time for abatement-year cohorts in 2005 and 2010 in Figures 4 and 5. These plots suggest increases. This dependent variable specifies output of the government sector, which includes public services, expenditures, and activity of enterprises which are "government agencies that. . . (sell) goods and services to the public."[36] While this variable is imprecise because it consists of federal, state, and local expenditures, the localized nature of the abatement coupled with the timing of the effect suggests that the observed directional increase could plausibly be driven by local government spending. This result suggests that data center abatements create a short-run local fiscal impact, which may reflect either an expansion of the local tax base or increased government output due to the costs of accommodating the new data center.

[36]U.S. Bureau of Economic Analysis (2022).

7.2 Total Revenue, Property Tax, and Bond/Debt Results

Using the TWFE regression from equation (1), I estimate the effects of data center tax abatements on county tax collections, property tax collections, and local government debt. I use county-level financial data from Annual Comprehensive Financial Reports to examine whether abatements lead to changes in fiscal outcomes in the one and two years following the abatement. These estimates provide preliminary evidence on whether data centers expand the local tax base or require debt-financed accommodations in the short run.

Table 4: Total Tax Revenue Effects

Model	**Coefficient**	**Std. Error**	**t-Statistic**	**Adj.** R^2
TWFE (Year 1 Effect)	0.01088	0.01162	0.93623	0.992
TWFE (Year 2 Effect)	0.00808	0.01178	0.68548	0.992

Note: The dependent variable is the natural logarithm of total county tax revenue. Both models include CompanyCountyEvent_ID and year-fixed effects. Standard errors are clustered at the CompanyCountyEvent_ID level. Year 1 Effect: N = 1,990 observations, 117 company-events, 42 counties. Year 2 Effect: N = 1,990 observations, 117 company-events, 42 counties. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

I estimate no statistically significant effects on total tax revenue collections in either the first or second year after abatement. Using 1,990 observations across 117 company-events in 42 counties, I estimate that total tax revenue increased by 1.1 percent in the year after abatement and 0.8 percent in the second year after abatement (Table 4). Both estimates are statistically insignificant, and these null results suggest that data center location does not generate detectable short-run increases in overall county tax collections, despite the potential for an expanded tax base from new commercial activity. The adjusted R^2 of 0.992 for both regressions indicates that the fixed effects explain nearly all variation in tax revenue, with the treatment effect explaining minimal additional variance. While the point estimates are positive, the lack of statistical significance means these results provide no evidence that tax abatements lead to increased total tax revenue in the immediate years following abatement.

Table 5: Property Tax Collections Effects

Model	**Coefficient**	**Std. Error**	**t-Statistic**	**Adj.** R^2
TWFE (Year 1 Effect)	-0.00558	0.02015	-0.27711	0.985
TWFE (Year 2 Effect)	0.00072	0.01783	0.04025	0.985

Note: The dependent variable is the natural logarithm of total county property tax collections. Both models include CompanyCountyEvent_ID and year-fixed effects. Standard errors are clustered at the CompanyCountyEvent_ID level. Year 1 Effect: N = 1,998 observations, 115 company-events, 41 counties. Year 2 Effect: N = 1,998 observations, 115 company-events, 41 counties. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Property tax collections similarly show no significant change following data center abatements. With 1,998 observations across 115 company-events in 41 counties, I estimate a 0.6 percent decrease in property tax collections in the first year after abatement and a negligible 0.07 percent increase in the second year (Table 5). Neither estimate approaches statistical significance. These results are particularly noteworthy given that property taxes represent the primary revenue source that data center abatements directly reduce. The near-zero coefficient in year two indicates no meaningful trend toward tax base expansion even as the data center becomes operational. This finding is consistent with data centers being capital-intensive facilities that generate limited spillover effects on local property values within a two-year window.

Table 6: Debt Borrowing Effects

Model	**Coefficient**	**Std. Error**	**t-Statistic**	**Adj.** R^2
TWFE (Year 1 Effect)	0.05522^{*}	0.03226	1.71166	0.962
TWFE (Year 2 Effect)	0.03022	0.03688	0.81940	0.962

Note: The dependent variable is the natural logarithm of county debt borrowing. Both models include CompanyCountyEvent_ID and year-fixed effects. Standard errors are clustered at the CompanyCountyEvent_ID level. Year 1 Effect: N = 862 observations, 71 company-events, 23 counties. Year 2 Effect: N = 862 observations, 71 company-events, 23 counties. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The debt borrowing results provide the most notable finding across the ACFR fiscal outcomes. Using 862 observations across 71 company-events in 23 counties, I estimate that county debt borrowing increased by 5.5 percent in the year after abatement, a result that is marginally significant at the p<0.10 level (Table 6). The effect attenuates to 3.0 percent in the second year and loses

statistical significance. Debt borrowing represents a broad measure of county borrowing that includes municipal bonds, short-term notes, bank loans, and other debt instruments. This effect suggests that counties may increase borrowing in the immediate aftermath of attracting a data center, potentially to finance infrastructure improvements or expanded services required to accommodate the new facility. The statistical significance in year one, followed by attenuation in year two, is consistent with front-loaded accommodation costs rather than sustained fiscal strain. However, the sample for this analysis is notably smaller than the tax revenue samples, with only 23 counties reporting debt borrowing data, which may introduce selection bias if counties that report detailed debt information differ systematically from those that do not.

Table 7: Bonded Debt Effects

Model	**Coefficient**	**Std. Error**	**t-Statistic**	**Adj.** R^2
TWFE (Year 1 Effect)	0.05368	0.03668	1.46360	0.948
TWFE (Year 2 Effect)	0.04073	0.03236	1.25873	0.948

Note: The dependent variable is the natural logarithm of county bonded debt. Both models include CompanyCountyEvent_ID and year-fixed effects. Standard errors are clustered at the CompanyCountyEvent_ID level. Year 1 Effect: N = 840 observations, 57 company-events, 16 counties. Year 2 Effect: N = 840 observations, 57 company-events, 16 counties. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Bonded debt shows a similar pattern to debt borrowing but does not achieve statistical significance. With 840 observations across 57 company-events in 16 counties, I estimate that bonded debt increased by 5.4 percent in the first year and 4.1 percent in the second year after abatement (Table 7). Bonded debt is a narrower measure than debt borrowing, capturing only formal, long-term municipal bonds issued in the public bond market rather than the full range of county borrowing instruments. The point estimates are similar in magnitude to the debt borrowing effects, but the smaller sample size and larger standard errors prevent the results from reaching conventional significance levels. The positive coefficient across both years, however, suggests a possible underlying positive effect that this analysis is underpowered to detect. The sample includes only 16 counties with available bonded debt data, representing the smallest sample among all fiscal outcomes and likely consisting of larger, more sophisticated counties that issue formal bonds in public markets.

Taken together, these fiscal results provide preliminary evidence that data center abatements generate short-run accommodation costs rather than immediate tax base expansion. The null effects on tax revenue and property tax collections indicate that counties do not experience increased tax collections in the two years following abatement, contradicting a simple narrative that abatements lead to fiscal windfalls from expanded commercial activity. The marginally significant increase in debt borrowing in year one, combined with the similar magnitude but insignificant increase in bonded debt, suggests that counties may finance data center-related infrastructure or services through borrowing rather than from an expanded revenue base. This pattern is consistent with the earlier BEA result showing approximately $200 per capita increases in government sector output—if government output increases but tax revenue does not, debt financing represents a plausible mechanism.

These results should be interpreted as preliminary, as the two-year window is not sufficient to observe longer-run effects such as sustained property tax base expansion or indirect economic development. Additionally, the sample period ends in 2020, prior to the rapid expansion of artificial intelligence applications that has dramatically increased data center demand and potentially altered their economic impacts. Future analysis will require more recent data to assess whether the fiscal patterns documented here persist in the AI era. I intend to expand this analysis by examining longer time horizons and heterogeneity by county characteristics, which may reveal whether the initial accommodation costs are offset by subsequent fiscal benefits as data centers become fully operational and the fraction of their non-abated value is fully capitalized into the local property tax base.

8. Conclusion

The BEA results indicate that data center tax abatements affect public finance in a non-trivial way. I find evidence of a positive increase in government sector output of approximately $200

per person in the year following the abatement, with a lingering effect in the subsequent year. This finding raised a critical question about the underlying mechanism: does the observed increase reflect increased local tax revenue from an expanded tax base, or does it reflect borrowing to finance accommodating costs such as infrastructure and expanded public services?

To investigate this mechanism, I collected county-level financial data from Annual Comprehensive Financial Reports. The preliminary fiscal results provide evidence favoring the accommodation costs hypothesis. I find no statistically significant effects on total tax revenue or property tax collections in either the first or second year following abatement. However, I estimate a marginally significant 5.5 percent increase in debt borrowing in the year after abatement ($p<0.10$), with a similar magnitude but insignificant increase in bonded debt. These patterns suggest that counties finance data center-related infrastructure through borrowing rather than from increased tax collections in the short run.

These fiscal results should be interpreted as preliminary. The two-year window is not sufficient to observe longer-run effects such as sustained property tax base expansion, and the sample period ends in 2020, prior to the AI-driven surge in data center demand. The debt variables also suffer from substantial sample attrition, with only 16-23 counties reporting debt data compared to 42 counties with tax revenue data. Future research should extend the time horizon, expand ACFR coverage for more recent years, and collect detailed information on specific tax types abated and local revenue sources. Nevertheless, the steadily increasing number of data centers being built with abatements highlights the importance of understanding their fiscal impacts as counties navigate the tension between attracting large-scale capital investment and maintaining fiscal sustainability.

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Tables

Table 8: Location of Tax Abatements by State and County for Sample

Alabama	1
Jackson	1
California	6
Los Angeles	1
Orange	1
Sacramento	3
Santa Clara	1
Colorado	1
Douglas	1
Georgia	1
Newton	1
Iowa	2
Dallas	1
Polk	1
Louisiana	1
Caddo	1
Michigan	1
Oakland	1
Minnesota	1
Itasca	1
Missouri	5
Atchison	1
Jackson	4
Nebraska	1
Douglas	1
Nevada	3
Clark	1
Storey	1
Washoe	1
New York	35
Erie	16
Niagara	1
Rensselaer	4
Rockland	14
North Carolina	3
Caldwell	1
Catawba	1
Cleveland	1
Ohio	1
Delaware	1
Oregon	9
Baker	2
Deschutes	1
Umatilla	1
Wasco	1
Washington	4
Pennsylvania	2
Allegheny	2
Texas	5
Dallas	1
Tarrant	2
Travis	2
Utah	1
Utah	1
Virginia	28
Chesterfield	2
Culpeper	1
Fauquier	3
Henrico	3
Loudoun	18
Mecklenburg	1
Washington	28
Douglas	9
Grant	14
King	5

Grand Total: 135 tax abatements **Note:** This table shows the number of tax abatements for data centers by state and county in the sample.

Table 9: Location of Tax Abatements by State and County for Annual Comprehensive Financial Reports Sample

Alabama	1
Jackson	1
California	5
Alameda	1
Los Angeles	1
Orange	1
Sacramento	1
Santa Clara	1
Colorado	1
Douglas	1
Georgia	2
Muscogee	1
Newton	1
Iowa	3
Dallas	1
Polk	2
Louisiana	1
Caddo	1
Michigan	1
Oakland	1
Minnesota	2
Carver	1
Itasca	1
Missouri	5
Atchison	1
Jackson	4
Nebraska	1
Douglas	1
Nevada	2
Storey	1
Washoe	1
New Jersey	2
Bergen	1
Hudson	1
New York	22
Erie	11
Niagara	1
Rensselaer	5
Rockland	6
North Carolina	3
Caldwell	1
Catawba	1
Cleveland	1
Oregon	8
Baker	2
Deschutes	1
Umatilla	1
Wasco	1
Washington	4
Pennsylvania	5
Allegheny	2
Chester, Philadelphia	1
Delaware	1
Westmoreland	1
Texas	5
Dallas	1
Harris	1
Tarrant	2
Travis	2
Utah	1
Utah	1
Virginia	29
Chesterfield	2
Culpeper	1
Fairfax	2
Fauquier	3
Henrico	3
Loudoun	8
Mecklenburg	1
Prince William	6
Virginia Beach	1
Washington	14
Douglas	5
Grant	5
King	4

Note: Total abatements = 116 across 54 counties in 20 states.

Table 10: Number of Tax Abatements by Year in Sample

Year	Number of Abatements
2003	1
2004	1
2005	3
2006	4
2007	2
2008	2
2009	5
2010	6
2011	15
2012	13
2013	19
2014	15
2015	9
2016	13
2017	9
2018	12
2019	3
2020	1

Note: This table reports the number of tax abatements recorded in each year of the sample. Abatement counts are based on the timing of the initial event year.

Table 11: Summary Statistics for Government Expenditures per Capita Regression Variables

Variable	n	Mean	Median	Min	Max	IQR	Count(1), Count(0)
GDPCap	2660	6.90	6.62	0.611	15.4	2.38	– , –
PostYear	2660	0.0496	–	–	–	–	132 , 2528
Post2Year	2660	0.0485	–	–	–	–	129 , 2531

Note: This table presents summary statistics for the regression analyzing government expenditures per capita. Binary indicators `PostYear` and `Post2Year` are shown with their counts of 1s and 0s; dashes (–) indicate statistics that are not meaningful for indicator variables.

Table 12: Sample Composition by Outcome Variable

Outcome Variable	Observations	Company-Events	Counties	Data Coverage
Total Tax Revenue	1,990	117	42	87%
Property Tax Collections	1,998	115	41	85%
Debt Borrowing	862	71	23	53%
Bonded Debt	840	57	16	42%
Full Sample	2,700	135	44	–

Note: This table shows the sample size for each dependent variable in the ACFR analysis. The full sample consists of 135 company-events across 44 counties from 2001-2020 (20 years $\times$ 135 events = 2,700 potential observations). Data Coverage shows the percentage of company-events with available data for each outcome variable. The reduced sample sizes for debt-related variables reflect limited reporting of debt borrowing and bonded debt information in county Annual Comprehensive Financial Reports, particularly among smaller counties.

Table 13: Summary Statistics for ACFR Regression Variables

Variable	N	Mean	Median	Min	Max	IQR
Log Total Tax Revenue	1,990	20.115	20.249	13.811	23.950	1.440
Log Property Tax Collections	1,998	19.439	19.827	12.824	23.667	1.603
Log Debt Borrowing	862	19.459	19.715	11.430	21.516	1.850
Log Bonded Debt	840	19.369	19.792	13.039	21.627	1.370

Note: This table presents summary statistics for the dependent variables used in the ACFR regression analysis. All variables are expressed as natural logarithms of county-level financial outcomes. IQR denotes the interquartile range (75th percentile - 25th percentile). The sample includes company-event-year observations from 2001-2020. Total Tax Revenue sample: 117 company-events, 42 counties. Property Tax Collections sample: 115 company-events, 41 counties. Debt Borrowing sample: 71 company-events, 23 counties. Bonded Debt sample: 57 company-events, 16 counties.

Figures

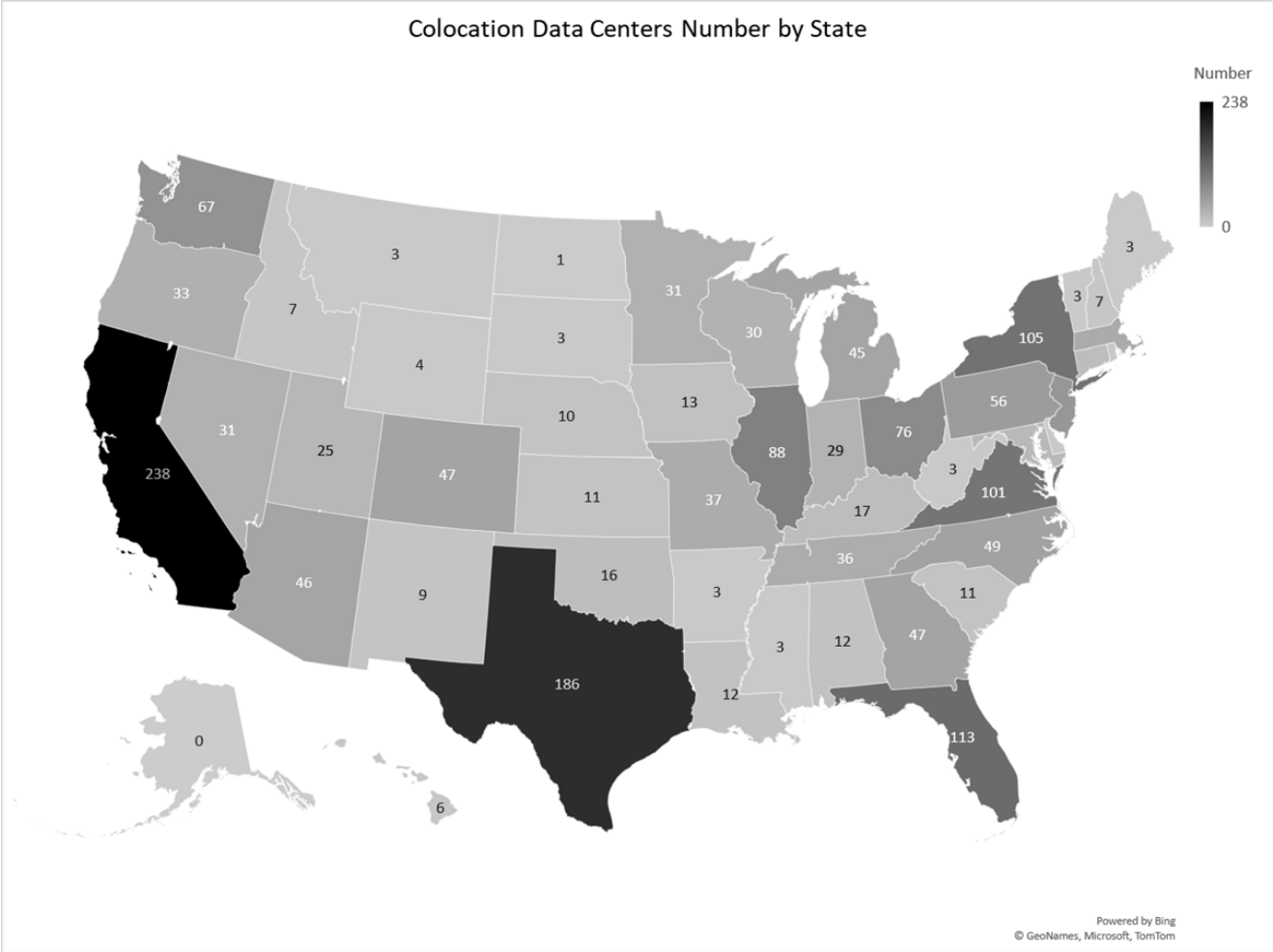

Figure 1: Number of Colocation Data Centers by State

Note: This figure indicates the location of the 1,829 Data Centers by state in 2022.

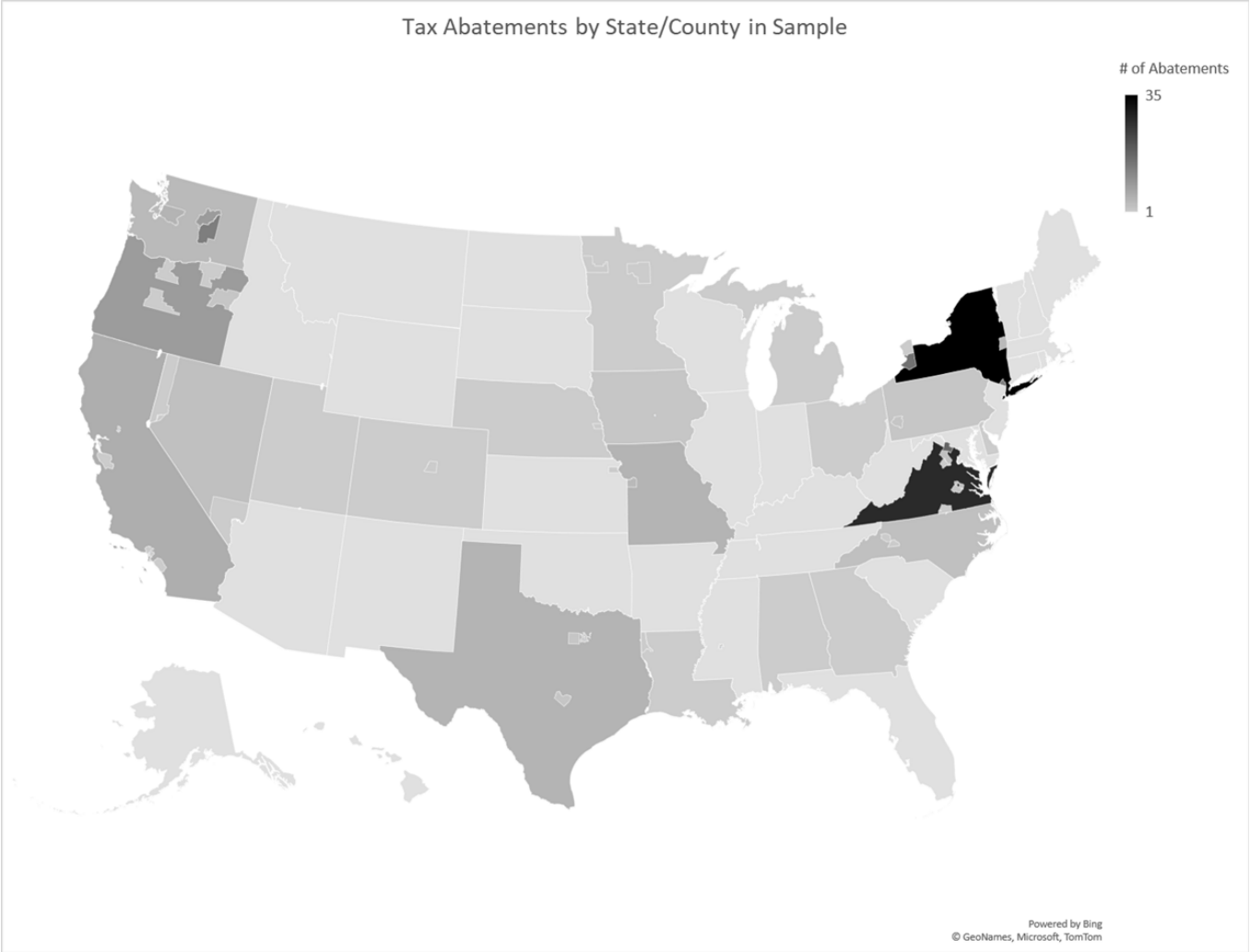

Figure 2: Map of Location of Tax Abatements for Data Centers in Sample

Note: This figure shows the geographic distribution of data centers that received tax abatements within the analytic sample. States included reflect those with verified incentives available during the study period.

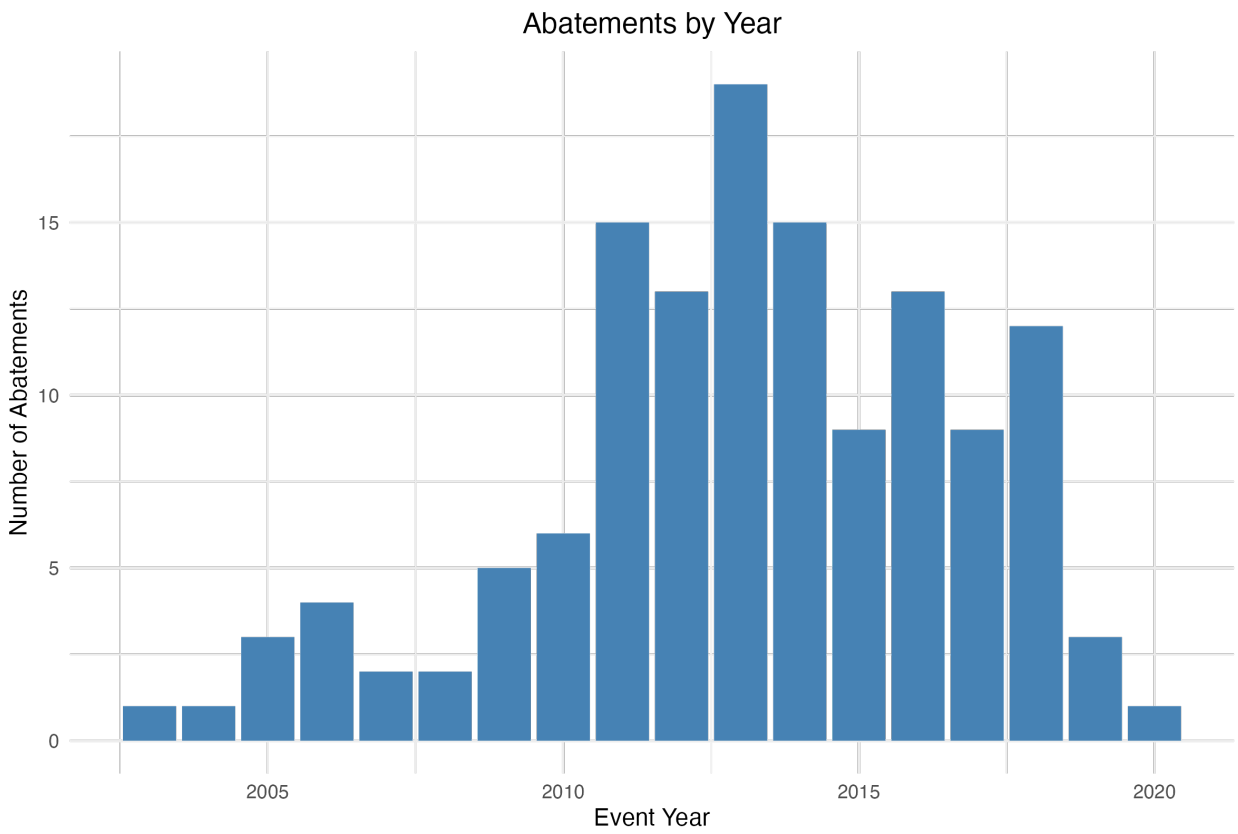

Figure 3: Abatements by Year

Note: This figure displays the number of unique data center tax abatements granted each year across counties in the sample. Each abatement is counted once per project per year using distinct combinations of `Company` and `EventYear`. The chart highlights the temporal distribution of abatements and provides context for identifying treatment exposure in the regression analysis.

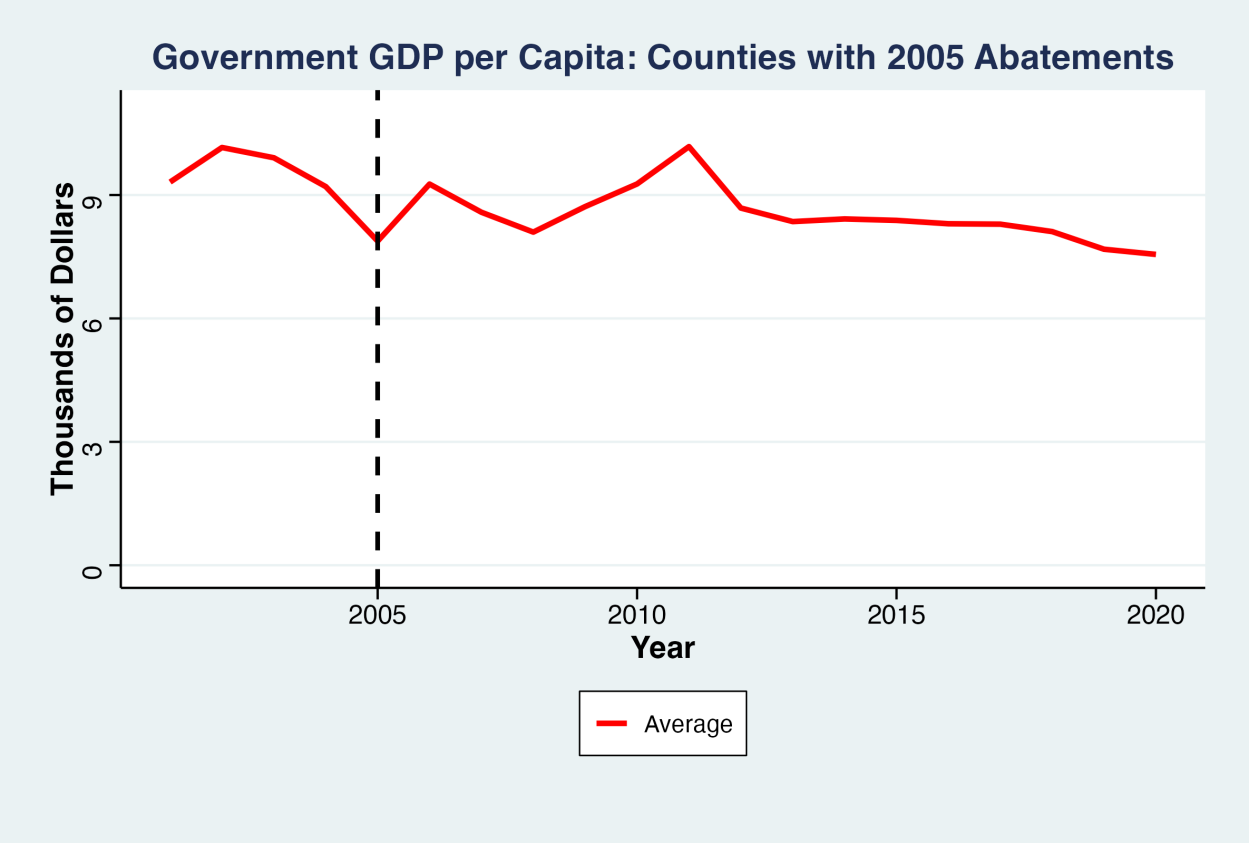

Figure 4: Average Government-GDP per Capita Across Counties in 2005

Note: This figure plots the average government GDP per capita (in thousands of U.S. dollars) across counties that received abatements in 2005. The vertical dashed line indicates the year of the abatement.

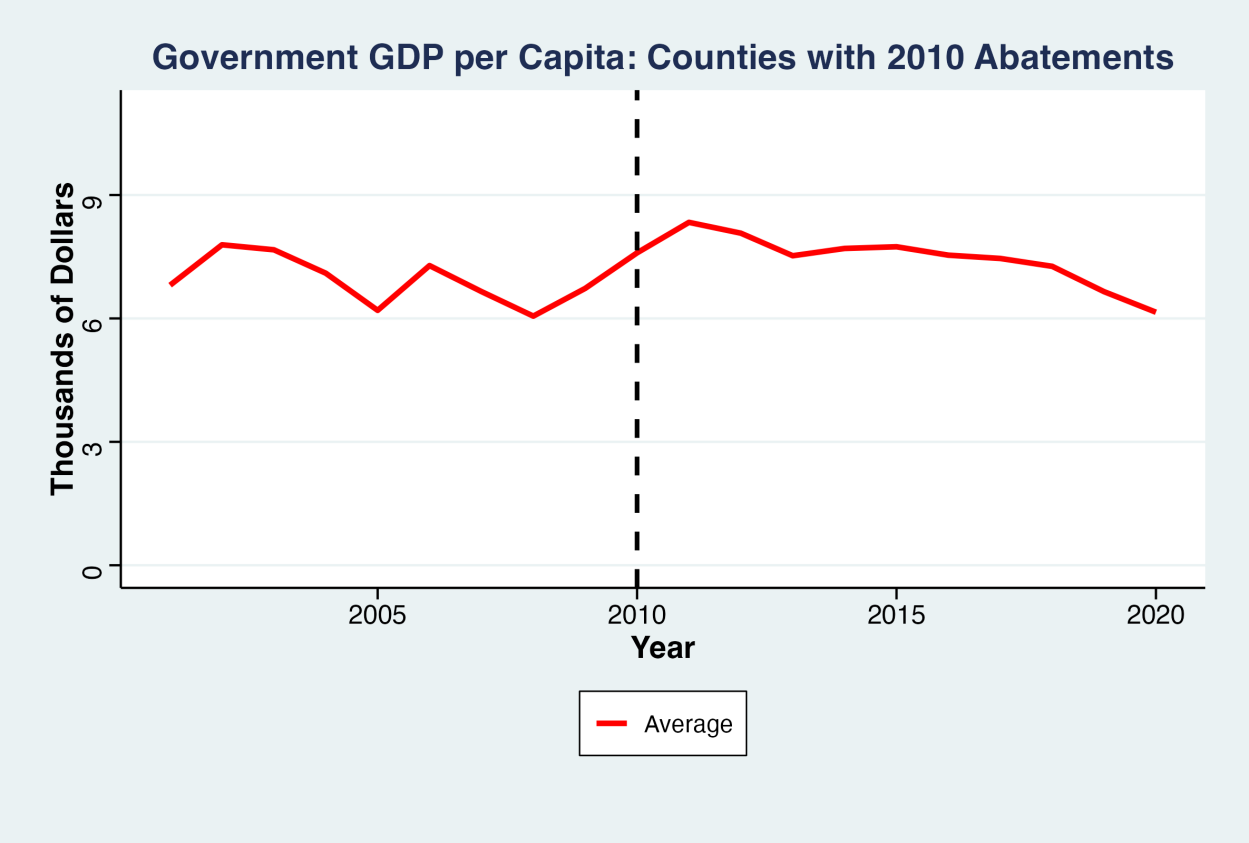

Figure 5: Average Government-GDP per Capita Across Counties in 2010

Note: This figure plots the average government GDP per capita (in thousands of U.S. dollars) across counties that received abatements in 2010. The vertical dashed line indicates the year of the abatement.